Program : **Diploma in Artificial Intelligence & Machine Learning**	
Course Code : **4347**	Course Title: **Machine Learning Lab**
Semester : **4**	Credits: **1.5**
Course Category: **Program Core**	
Periods per week: **3 (L:0 T:0 P:3)**	Periods per semester: **45**

Course Objectives:

Students will be able to make use of Data sets in implementing the machine learning algorithms and also to implement the machine learning concepts and algorithms in Python.

Course Prerequisites:

Topic	Course name	Semester
Basic Mathematics	Mathematics I	1
	Mathematics II	2
Basics of Python programming	Introduction to IT systems Lab	1
Programming Concepts.	Programming in C	2
	Programming in Python Lab	3

Course Outcomes :

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Installation of Anaconda and familiarization of various packages	12	Applying
CO2	Develop solution for Supervised learning	15	Applying
CO3	Develop solution for Unsupervised learning	9	Applying
CO4	Open Ended Experiments	6	Applying
	Lab Exam	3	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2	3			3			
CO3	3			3			
CO4	3	3	3	3	3	3	3

3-Strongly mapped, 2-Moderately mapped , 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Installation of Anaconda and familiarization of various packages**		
M1.01	Practice installation of anaconda	6	Applying
M1.02	Installation and familiarization of various packages - NumPy, Pandas, sklearn, matlibplot etc.		
M1.03	Practice import and export data using library functions by identifying any five synthetic data sets from various online resources. Data sets can be taken from standard repositories (eg:https://archive.ics.uci.edu/ml/datasets.html) or constructed by the students. Use appropriate data sets, for modelling the below problems, wherever necessary		

M1.04	Write a Python program to do the following operations: Library: NumPy a) Create multi-dimensional arrays and find its shape and dimension b) Create a matrix full of zeros and ones c) Reshape and flatten data in the array d) Append data vertically and horizontally e) Apply indexing and slicing on array f) Use statistical functions on array - Min, Max, Mean, Median and Standard Deviation	3	Applying
M1.05	Implement data preprocessing (eg: Normalization, Processing missing data)	3	Applying
CO2	**Develop solution for Supervised learning**		
M2.01	Implement k-nearest neighbours classification on datasets	3	Applying
M2.02	Implement linear regression on datasets	3	Applying
M2.03	Implement decision tree	3	Applying
M2.04	Implement Random Forest	3	Applying
M2.05	Implement Naïve Bayes	3	Applying
	Lab Exam – I	1.5	
CO3	**Develop solutions for Un-Supervised Learning Problems**		
M3.01	Implement dimensionality reduction using Principal Component Analysis (PCA)	3	Applying
M3.02	Implement K-means clustering	3	Applying
M3.03	Implement Mean Shift clustering	3	Applying
CO4	**Open Ended Experiments****	6	Applying
	Lab Exam – II	1.5	

**** - Suggested Open Ended Experiments**

(Not for End Semester Examination but compulsory to be included in Continuous Internal Evaluation. Students can do open ended experiments as a group of 2-3. There is no duplication in experiments between groups)

1. Coronavirus break down
2. Cancer subtype identification
3. Prediction on House prices
4. Prediction of survival on Titanic

Text / Reference

T/R	Book Title/Author
T1	Kevin P. Murphy. Machine Learning: A Probabilistic Perspective, MIT Press 2012.
R1	Ethem Alpaydin, Introduction to Machine Learning, 2nd edition, MIT Press 2010
R2	Mitchell M., T., Machine Learning, McGraw Hill (1997) 1stEdition.

Online Resources

Sl.No	Website Link
1	https://developers.google.com/machine-learning/crash-course/
2	https://studyglance.in/labprograms/mllabprograms.php
3	https://machinehack.com/